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AURORA: A Regime-Switching Forecasting Architecture with Verified Causal Integrity and Calibrated Uncertainty for Financial Time Series

A Research Proposal Submitted to the International Academic Research Competition (IARCO) 2026

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Research Topic	Artificial Intelligence (Machine Learning for Financial Forecasting)
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Submission Date	21/09/2026

Research Focus: Designing and evaluating a seven-layer regime-switching architecture that produces accurate, seed-stable, and calibrated forecasts of financial time series under a pre-registered walk-forward protocol whose freedom from look-ahead bias is enforced by an executable test.

Keywords: regime switching; mixture of experts; conformal prediction; hidden Markov models; financial time-series forecasting; look-ahead bias; walk-forward validation

Abstract

Deep learning now dominates financial time-series forecasting, yet its comparative evidence base is contradictory: model rankings reverse across horizons, across market conditions, and across studies. This proposal argues that the inconsistency is structural, and addresses it on two fronts. It proposes Aurora, a seven-layer regime-switching architecture that conditions feature construction, forecast combination, uncertainty quantification, and the decision to act on an online-estimated market state; and a protocol in which the claims usually asserted in this literature are instead tested: freedom from look-ahead by a bit-exact truncation-invariance assertion, stability by across-seed dispersion reported as a first-class result, and significance by correction across the entire comparison family. A completed pilot on thirty years of S&P 500 data establishes feasibility.

1. Introduction

Financial time-series forecasting is a dominant machine learning application, with reviews documenting the rapid adoption of recurrent, attentional, and generative architectures, [2], and nonlinear methods finding genuine predictive content in asset pricing. Conversely, the **weak form of the efficient market hypothesis** argues past prices lack exploitable signals, and equity-premium predictors suffer from well-documented out-of-sample fragility. Comparative evidence sits uneasily between these positions: results reverse across horizons, market conditions, and studies, with no single model family establishing a durable advantage.

This proposal argues this inconsistency reflects **structure rather than noise**. Markets alternate between distinct statistical states: the persistent low-volatility, positive-drift condition and the high-volatility, negative-drift condition formalised by Hamilton [6] and shown by Ang and Bekaert to be allocation-relevant. Because model families carry different inductive biases, they are not equally matched to every state or horizon. Studies averaging over these variables measure an average of conditional advantages, misleadingly reporting it as an unconditional one.

A second problem compounds the first: much reported evidence displays **fragility invisible in published numbers**. Pipelines joining macroeconomic series on reference dates instead of release dates silently grant models future information, [9]. Architectures whose long-horizon error varies by fifteen index points across random seeds are often compared at a single seed. Furthermore, studies reporting the best of several models across horizons adjust significance within tables rather than across the whole family, [11]. While none of these defects raises a explicit error, each inflates results. A third gap is practical: a point estimate without an honest uncertainty statement or a principled basis for declining to act cannot support a decision.

2. Research Questions and Hypotheses

The independent variable is whether feature construction, combination, calibration, and execution are conditioned on a causally estimated regime posterior. The dependent variables are accuracy (RMSE, MAE, MAPE, directional accuracy), across-seed dispersion, empirical interval coverage against a nominal target, and risk-adjusted return net of modelled costs.

RQ1. Does conditioning a forecast combiner on an online-estimated regime posterior improve accuracy over regime-agnostic equal-weight combination, at horizons of one, five, and twenty trading days?

RQ2. Can prediction uncertainty be made state-dependent while retaining near-nominal empirical coverage, by conditioning split-conformal calibration on the filtered regime?

RQ3. How much of the variation in reported comparative results is attributable to random initialisation rather than to architecture?

RQ4. Can freedom from look-ahead bias be established by executable test rather than by assertion?

RQ5. Does abstention on a calibrated criterion convert a forecast into a positive risk-adjusted return after realistic transaction costs?

The hypotheses are: (H1) regime-conditioned routing outperforms regime-agnostic combination at every horizon, because expert skill is conditional on market state [6], [7]; (H2) conditioning the calibration set on the filtered regime improves the exchangeability of conformity scores within it and so yields near-nominal coverage with regime-adaptive width [12], [13]; (H3) joint training under a constraint on prediction magnitude reduces across-seed dispersion by roughly an order of magnitude relative to an unconstrained adversarial forecaster; (H4) a forward-only regime estimator satisfies exact truncation invariance, so any upstream look-ahead causes a mechanical test failure; and (H5) calibrated abstention improves net Sharpe ratio at the horizon where costs bind hardest.

3. Literature Review

Three strands complicate a blanket verdict of unpredictability. The first is the microstructure of returns: Cont catalogues heavy tails, near-absence of linear autocorrelation in returns coupled with strong, slowly decaying autocorrelation in absolute returns, and a negative return-volatility correlation [14]. The second is regime structure, formalised by Hamilton's Markov-switching model [6] and shown to change optimal allocation materially [7]. The third is the machine-learning asset-pricing literature, which finds modest but genuine conditional predictability in nonlinear interactions [3].

On the modelling side, the Long Short-Term Memory network overcomes the vanishing-gradient problem that prevents plain recurrent networks from learning long-range dependencies [15], with Fischer and Krauss providing an influential demonstration on S&P 500 constituents [16]. The Transformer replaces recurrence with self-attention [17]; adversarial networks pose learning as a minimax game [18] and were adapted to sequential financial data by TimeGAN [19], with the Wasserstein formulation mitigating training instability [20]; and gradient-boosted trees remain competitive with deep architectures on tabular financial predictors [3], [21]. The head-to-head evidence against classical benchmarks is mixed: Siami-Namini et al. report large error reductions for LSTM over ARIMA [22], while the forecasting-competition literature finds that across large collections of series, machine-learning methods underperform simple statistical benchmarks [23]. This heterogeneity is the puzzle a conditioned design resolves.That combining forecasts improves accuracy is among the oldest robust findings in the field [24], and Timmermann explains why: combination diversifies misspecification and estimation error, and equal weighting is famously hard to beat [25]. This is why a combiner that beats its own members is not evidence of anything interesting, and why the decisive control here is regime-agnostic combination rather than the individual experts. Mixture-of-experts supplies the formalism for conditional combination [26]; the departure proposed is to drive the gate from an externally estimated, economically interpretable latent state rather than an unrestricted function of the features.

Two further literatures inform the protocol. Point-in-time discipline addresses the fact that macroeconomic aggregates are released with a lag and then revised, so a naive join grants a backtest information it could not have had [8], [9]. Conformal prediction constructs prediction sets with finite-sample marginal coverage under exchangeability alone, with no assumption on the model [12], [13]; because financial series are not exchangeable, methods valid under distribution shift are an active area [27]. The multiple-testing literature supplies the instruments for honest reporting: the Diebold-Mariano test [28] with the Harvey-Leybourne-Newbold correction [29], the Deflated Sharpe Ratio [30], and the Probability of Backtest Overfitting [11]. The resulting gap is threefold: the comparative literature is contradictory in a way consistent with skill being conditional; the few studies conditioning on market state rarely combine regime stratification, multi-horizon evaluation, and family-wide significance testing in one design; and essentially none closes the loop by introducing a method that uses the measured conditionality and testing it against the regime-agnostic control that would reveal whether regime information adds anything beyond averaging.

4. Proposed Methodology

Approach and data. The study is quantitative and computational, combining an exploratory phase (characterising regime structure and horizon-dependent skill), a predictive phase (developing the architecture), and an applied phase (converting forecasts into cost-aware decisions), all in one reproducible Python harness. The target is the S&P 500 composite index at daily close over 1990-2019 (n = 7,303 trading days), forecast as the h-day log return in (1). The window is fixed before modelling and for a stated reason: it contains two complete bear markets, which a regime-switching study cannot do without, and ends before the COVID-19 dislocation, which would otherwise sit in the evaluation window as a single dominant outlier. Exogenous series are the CBOE volatility index, WTI crude, a trade-weighted dollar index, and the Shiller valuation dataset. The feature set comprises 69 predictors

in five families (returns and momentum, realised volatility with range estimators, technical indicators, cross-asset levels, and valuation and rate variables), standardised using training-fold statistics only.

$$\rho_{t,h} = \log P_{t+h} - \log P_t, \qquad \hat{P}_{t+h} = P_t \exp(\hat{\rho}_{t,h}) \tag{1}$$

Architecture. Aurora is a seven-layer stack, each layer carrying an independent on/off switch so that the ablation is a clean one-factor-at-a-time design.

L0. Availability graph. Every feature carries release lag, revision lag, event offset, and release hour; a value is masked from a forecast made at time s unless it was knowable before s. For feature j with release lag ℓ_j, event offset o_j and release hour η_j, first availability is given by (2); a forecast at s sees $X_{\tau,j}$ only if $a_j(\tau) \le s$, and every non-daily family must satisfy $v_j = o_j + \ell_j > 0$. Folds are purged and embargoed by at least the horizon, since a target at index i depends on prices through $i + h$; for a test block starting at i_0 with embargo e, the admissible training set is (3).

$$a_j(\tau) = \tau + o_j + \ell_j + \eta_j \tag{2}$$

$$\mathcal{T} = \{\, i : i < i_0 - \max(e, h) \,\} \tag{3}$$

L1. Online regime estimation. A three-state Gaussian hidden Markov model over realised return, volatility, and drawdown supplies the filtered posterior only, computed by the forward recursion (4) to (5) with transition matrix A and emission densities b_k; transition risk (6) feeds the gate in (11). The distinction from the smoothed posterior is the most important causal detail in the system: the smoothed quantity conditions on the whole sample, importing the future into the label.

$$\alpha_t(k) = b_k(y_t) \sum_m \alpha_{t-1}(m) A_{mk} \tag{4}$$

$$\gamma_t(k) = \frac{\alpha_t(k)}{\sum_m \alpha_t(m)} \tag{5}$$

$$r_t = 1 - \sum_k \gamma_t(k) A_{kk} \tag{6}$$

L2. Regime-conditioned interactions. A bounded, rank-one Fourier term lets a predictor's representation depend multiplicatively on the regime posterior, supplying the interaction directly rather than making the network discover it.

L3. Bounded-move penalty. The forecaster trains end to end with a differentiable filter under the objective (7), where B_t is a rolling quantile of realised absolute move estimated on training data only, the financial analogue of constrained learning [31]. The penalty applies to the prediction, not the error: its argument in (7) is the magnitude of the predicted move, not of the forecast error. Penalising error merely reweights the loss, whereas penalising implausible predictions restricts the hypothesis class.

$$\mathcal{L} = \frac{1}{n} \sum_t (\hat{\rho}_t - \rho_t)^2 + \beta \frac{1}{n} \sum_t \max(0,\ |\hat{\rho}_t| - B_t) \tag{7}$$

L4a. Out-of-fold router. A softmax gate on γ_t produces weights w_t over expert forecasts p_t, fitted on out-of-fold predictions by minimising (8) with shrinkage toward the uniform vector u, subject to $\Sigma_i w_i = 1$, $w_i \le w_{\max}$ and $\|w_t - w_{t-1}\|_1 \le \delta$. The $1/\mathrm{Var}(\rho)$ factor is required, since unnormalised squared error on daily log returns is of order 10^{-4}. Fitting in-fold would let an overfitting expert attract weight it does not deserve.

$$J(\Theta) = \frac{1}{\mathrm{Var}(\rho)} \frac{1}{n} \sum_t (p_t^{\top} w_t - \rho_t)^2 + \lambda\, \mathrm{KL}(w_t \,\|\, u) + \mu \|\Theta\|^2 \tag{8}$$

L4b. Regime conformal intervals. Split-conformal calibration performed separately within each filtered regime, using residuals R_i in the calibration subset C_k of size n_k at miscoverage level ε with the finite-sample quantile correction (9), and a pooled fallback for sparse states.

$$\hat{q}_k = \text{Quantile}\left(\{R_i\}_{i \in C_k};\ \frac{\lceil (1-\varepsilon)(n_k+1) \rceil}{n_k} \right) \tag{9}$$

L5. Cost-aware abstention. Round-trip cost combines a proportional spread s with a square-root impact term (10) [32]; a position is taken only when expected edge exceeds cost by a multiple κ of the conformal half-width, with further gates on regime confidence $\pi_t = \max_k \gamma_t(k)$, transition risk r_t, data freshness φ_t, and liquidity ψ_t, as in (11).

$$c(v) = s + \zeta \sigma \sqrt{v/V} \tag{10}$$

$$\text{act} \Leftrightarrow (|\hat{\rho}_t| - c > \kappa \hat{q}_k) \wedge (\pi_t > \pi_{\min}) \wedge (r_t < r_{\max}) \wedge (\varphi_t > \varphi_{\min}) \wedge (\psi_t > \psi_{\min}) \tag{11}$$

L6. Verification artefacts. The final layer emits evidence rather than forecasts: test statistics, adjusted p-values, overfitting diagnostics, and the test suite below.

Evaluation protocol. Five expanding-window walk-forward folds with a minimum ten-year training length; horizons of one, five, and twenty days fixed in advance; three seeds per stochastic model. Baselines are a random walk with drift, ARIMA selected by BIC per fold, an attention-pooled LSTM, a Transformer encoder, a TimeGAN-style adversarial forecaster, gradient-boosted trees, an equal-weight ensemble, and a hindsight oracle retained only as an attainability bound. Significance uses Diebold-Mariano tests [28] with the Harvey-Leybourne-Newbold correction [29] and Newey-West HAC variance at lag $h - 1$, as in (12), Holm-Bonferroni adjusted across the entire family of comparisons rather than within individual tables. The trial count for the Deflated Sharpe Ratio [30] is declared before any dependent result is reported, and the Probability of Backtest Overfitting [11] is estimated by combinatorially symmetric cross-validation.

$$\text{DM}^{*} = \frac{\bar{d}}{\sqrt{\hat{V}(\bar{d})}} \sqrt{\frac{n + 1 - 2h + h(h-1)/n}{n}} \sim t_{n-1} \tag{12}$$

Verification. The central causal claim is enforced by a truncation-invariance assertion: the filtered posterior (5) computed on a series truncated at t must equal, to the last bit, the posterior computed on the full series and then truncated. Exactness is attainable because the forward recursion (4) at step t reads only the preceding forward variable and the current observation. Any implementation that standardises over the full sample, or passes a smoothed posterior downstream, fails. Thirteen further tests cover the purge-and-embargo rule, a strictly positive net availability lag for every non-daily family, bounded-move loss semantics, router projection geometry and objective scaling, and per-regime conformal coverage on synthetic data.

5. Preliminary Results and Feasibility

A complete pilot has been built and run on the sample above, establishing feasibility and providing the evidence base from which the proposed extensions follow. Table 1 reports accuracy pooled across five folds, three horizons, and three seeds.

Forecaster	RMSE	MAE	Dir. acc.	Seed s.d. (h=20)
ARIMA (classical ref.)	37.72	27.14	0.570	0.00
Attention-pooled LSTM	41.83	30.22	0.502	7.23
Transformer encoder	41.21	29.89	0.504	3.00
TimeGAN-style adversarial	39.50	28.01	0.567	15.44
Gradient-boosted trees	42.65	30.73	0.522	0.42
Equal-weight ensemble	38.88	27.89	0.541	1.53
Aurora (proposed)	38.20	27.91	0.520	2.02

Table 1. Aggregate out-of-sample accuracy. RMSE and MAE in index points; seed s.d. is the standard deviation of $h = 20$ RMSE across three initialisations. ARIMA shows 0.00 because it is deterministic, not because the value is missing.

Aurora leads the learned pool on every error metric, with margins surviving Holm-Bonferroni adjustment across the entire comparison family at $p < 0.05$, and improves on the regime-agnostic equal-weight control at every horizon, the contrast that isolates the regime signal from the general benefit of combination. Its across-seed dispersion at twenty days is roughly eight times smaller than the adversarial baseline's. Its regime-conditioned intervals attain 90.3% and 90.0% coverage against a 90% target at $h = 1$ and $h = 5$, with half-widths adapting across market states by a factor of 1.88. A layer ablation isolates each mechanism: the bounded-move penalty contributes most to accuracy, the router second, and replacing the regime conformal layer with an unconditional interval costs 11.7 percentage points of coverage.

6. Expected Outcomes

The programme is expected to deliver: **(i)** An extension beyond a single index to single-name equities, a second developed market, and a less efficient emerging market, testing whether regime conditionality strengthens where efficiency is weaker. **(ii)** An exogenous-sentiment variant on the post-2011 period, treating the shortened sample as an explicit trade-off rather than substituting a back-propagated proxy. **(iii)** Selection of the state count by information criterion rather than economic prior, preventing a misspecified count from attenuating regime-conditional results. **(iv)** Conformal methods valid under dependence, addressing the coverage shortfall at twenty days. **(v)** A released, reproducible harness ensuring every reported number can be re-derived rather than taken on trust.

7. Risk Management and Limitations

Four mitigated risks are identified: **Data quality:** Exogenous series are revised after release, so all non-daily families carry an asserted strictly positive net availability lag. A pilot defect involving a negative event offset was caught by this assertion. **Overfitting to the search process:** Trial counts are declared in advance, the Probability of Backtest Overfitting is reported, and execution hyper-parameters are selected on validation folds strictly preceding each test block. **Instability:** Every stochastic model runs at multiple seeds with dispersion reported. **Non-stationarity beyond regimes:** Discrete switching cannot capture structural breaks in microstructure or policy, meaning results should not be extrapolated without re-estimation.

8. Project Practicalities, Timeline, and Post-Program Plan

Because the pilot harness and evaluation infrastructure already exist, the eight-month programme begins from a working system: **Months 1-2:** Assemble and validate multi-asset and multi-market datasets under the existing availability-graph discipline. **Months 3-4:** Re-estimate and evaluate on each new market, using state-count selection by information criterion. **Months 5-6:** Integrate exogenous sentiment on the restricted post-2011 sample and implement conformal calibration valid under dependence. **Months 7-8:** Complete the ablation and significance analysis across all markets, prepare the public release, and write up.

9. Conclusion

Comparative literature reports contradictory rankings because predictive skill in this domain is conditional, but the evaluation protocols are not. This proposal addresses both halves with one design: an architecture conditioning feature construction, combination, calibration, and execution on a causally estimated market state, paired with a protocol where typically asserted claims (no look-ahead, stable performance, honest uncertainty) are explicitly tested and reported-including where they fail. Its most transferable contribution may be the smallest: a test converting an unfalsifiable causal claim into a falsifiable one.

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